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# FPGA-Based Adaptive High-Precision Hydrogen Sensing System

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## INTRODUCTION

For hydrogen-leak monitoring, this FPGA-based platform digitizes the resonance response of a hydrogen-sensitive microcavity and integrates digital laser locking, real-time beat readout, thermal management and UDP host visualization.



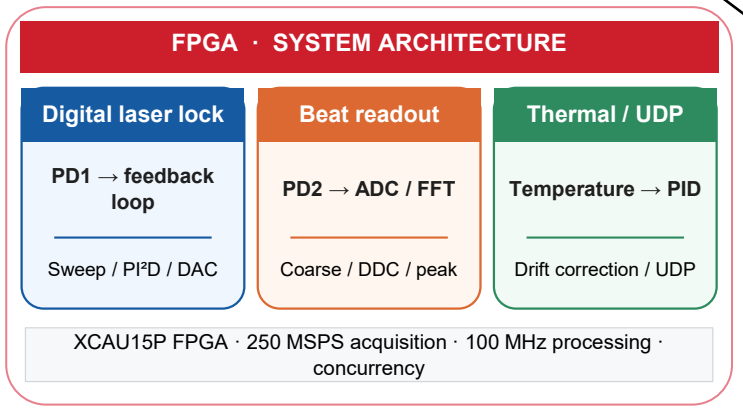
Sensing cavity and GW-5000 gas mixer

### Three key technical challenges

**1 Stable laser-frequency locking**
Analog servos are difficult to reconfigure; resonance drift and feedback noise impair long-term slope locking.

**2 Wideband, real-time beat readout**
Manual spectrum-analyzer readout limits online use; fast wideband capture must resolve small frequency changes.

**3 Optoelectronic integration**
Separate servos, analyzers and thermal units add size and wiring, while multi-rate control and drift remain challenging.



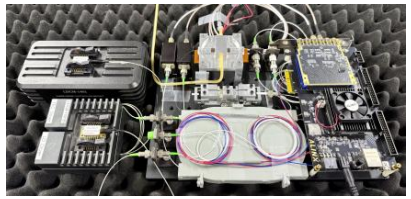

FPGA architecture and three functions

## CREATIVE DESIGN

**1 NEW CAPABILITY | Digital laser servo**
FPGA finds the resonance slope; PI²D and DAC close the feedback loop, with automatic reacquisition.

**2 NEW TECHNIQUE | Real-time beat readout**
250-MSPS capture and 4096-point FFT locate the band; DDC/CIC, Hann window and interpolation refine the peak.

**3 NEW ARCHITECTURE | Integrated FPGA**
Concurrent locking, FFT and thermal PID with clock-domain handshakes, drift correction and UDP host exchange.



*FPGA hardware and optical interfaces*

## RESULT

### Evidence across three core functions

**① DIGITAL LOCKING | Digital validation**

**456 → 2.26–2.64**

Error RMS / ADC codes

Four slope and initial-error cases; scanning, acquisition, locking and reacquisition.

**② BEAT READOUT | simulation + bench**

**5 kHz – 25 MHz**

Selected tested frequencies

Crossover offsets ≤3 Hz at tested points; 3.76–4.00 ppm for 1–25 MHz; ~4.780-ms updates.

**③ FPGA INTEGRATION | verification**

**17 checks · WNS +0.006 ns**

Joint simulation / implementation

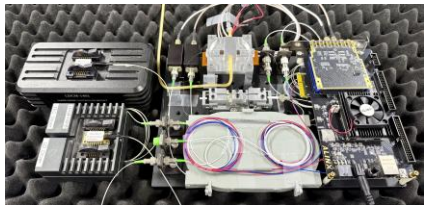
No setup violations (constrained paths); PID thermal model 25→40 °C; UDP dual-view host UI.

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# 基于FPGA的自适应高精度氢气传感系统

高穆妍、马俪娜、舒夏泽

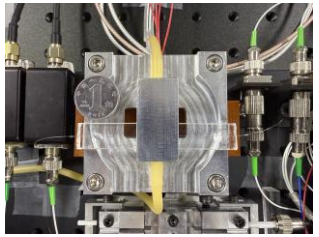
电子科技大学



*FPGA系统实物连接*

## 背景介绍

面向氢能储运的泄漏监测，本作品以FPGA为核心，将氢敏微腔的谐振频移转化为数字测量结果，并集成激光全数字锁定、拍频实时测频、温度管理与UDP上位机交互。





**氢敏微腔与自动配气仪**

### 三项关键技术难点

**1 激光频率稳定锁定难**

传统模拟伺服器结构封闭、参数调节不灵活；谐振漂移与反馈噪声影响长期稳定锁定。

**2 宽频拍频实时分析难**

频谱仪人工读数不利于在线监测；宽频快速采集与细小频移分辨难以同时兼顾。

**3 光电系统一体集成难**

伺服器、频谱仪和温控设备分立，体积及接线复杂；多速率协同与温漂抑制困难。

**FPGA • 系统总体架构**

| 数字激光锁定 | 实时拍频测频 | 温控与数据交互 |
|---|---|---|
| PD1光强 → 反馈算法 | PD2拍频 → ADC / FFT | 温度采集 → PID / PWM |
| 寻峰 / PI²D / DAC | 粗测 / DDC / 插值 | 温漂修正 / UDP |

XCAU15P FPGA：250 MSPS采样 · 100 MHz处理 · 并行运行

**FPGA三大核心功能与信号链路**

**1 新能力 | 全数字激光伺服锁定**

FPGA自动寻峰、捕获谐振斜边；PI²D经DAC反馈调谐，并在失锁后重搜。

**2 新技术 | 宽频拍频实时测频**

250 MSPS采样和4096点FFT粗定位；DDC/CIC抽取、Hann窗与插值实现精细读频。

**3 新架构 | 光电一体化集成**

锁定、测频和温控PID并行运行；跨时钟域交接、温漂修正与UDP上位机交互。

CREATIVE DESIGN

RESULT

### 三项核心功能的验证结果

**① 激光全数字锁定 | 数字闭环验证**

**456 → 2.26–2.64**

误差RMS / ADC码

4类斜边及初始偏差工况；自动扫描、捕获、锁定与失锁重搜。

**② 拍频实时测频 | 仿真及板测**

**5 kHz – 25 MHz**

已测离散频点范围

交叠区已测点偏差≤3 Hz；1–25 MHz相对偏差3.76–4.00 ppm；更新约4.780 ms。

**③ 系统一体集成 | 联合验证**

**17项检查 • WNS +0.006 ns**

联合数字时序实现

0建立违例（既定约束）；PID模型25→40 ℃；UDP双窗口上位机。